

Goals: Create normalized relational models that adheres to the 3NF.

Delivery: Make a copy of this document and fill in the provided sections for the problem you are working on. Turn in this assignment in Canvas as a PDF file, named as HW4_<problemID>_<studentID>.pdf, where <problemID> is the number of the problem you're working on and <studentID> is your student ID number (only numbers, without the 'a'). Submit the .pdf file as your solution for this assignment.

Individual assignment: do not work with others.

Rubrics: the rubrics are provided in Canvas.

General Instructions:

This assignment is divided into two parts. Firstly, you will convert a denormalized schema into one or more schemas in the 3NF. Then, you will draw a diagram that represents the schemas you modeled. The following sections must be filled:

1. **Denormalized schema:** reproduce here the denormalized schema, which is provided to you in The Problem section of this assignment in Canvas.
2. **Normalized schemas:** your version of the denormalized schemas that contains one or more relations, all of them in the 3NF.
3. **Relational diagram:** draw a diagram to represent your model in the 3NF using your preferred tool. Create an image of your diagram and add to this section.

Don't forget to generate a PDF of this document before submission. Other file extensions won't be accepted.

1. Denormalized schema

Zoo(animal_id(PK), animal_name, animal_species, animal_birthdate, animal_habitat, exhibit_id(PK), exhibit_name, exhibit_capacity, visitor_id, visitor_name, visitor_email, ticket_price, ticket_date, ticket_owner)

2. Normalized schemas

Relation 1: Exhibit(exhibit_id(PK), exhibit_name, exhibit_capacity)

Relation 2: Animal(animal_id(PK), animal_name, animal_species, animal_birthdate, animal_habitat, exhibit_id(FK))

Relation 3: Visitor(visitor_id(PK), visitor_name, visitor_email)

Relation 4: Ticket(visitor_id(PK, FK), ticket_id(PK), ticket_price, ticket_date, ticket_owner)

3. Relational diagram

